Market Ecology Simulator

Quick Start Guide: what the project is and how to work with it

1 · What the project is

An agent-based model of an equity market. It does not forecast prices. It asks **where market behaviour comes from**.

615 agents each follow a simple rule. None knows the true value of an asset, none knows what the others will do next, and **nowhere in the code is there an equation for the price**. Every trade is one order meeting another: a continuous double-auction order book with price-time priority, partial fills and expiry. Bubbles, crashes, volatility clusters and forced-selling cascades emerge from this interaction, or they are not there.

The question

Real return series share a handful of statistical properties that hold across asset classes and centuries. **None of them is built into this model.** Does a market made only of simple, heterogeneous rules meet them anyway? The criteria are executable checks in `mes/metrics.py`, fixed *before* anything was measured. That is the difference between a result and a story told afterwards.

Criterion	Requirement	Measured	Met
Volatility clustering	mean ACF of |returns| > 0.05	`0.217`	`25 / 25`
Non-normality	Jarque-Bera p < 0.05	`0.0002`	`25 / 25`
Fat tails	excess kurtosis > 1	`5.39`	`24 / 25`
Tail index	Hill estimator between 2 and 6	`4.38`	`22 / 25`
No autocorrelation	|ACF(1)| < 0.1	`−0.223`	`8 / 25`
Volume coupling	corr(volume, |return|) > 0.1	`−0.072`	`3 / 25`

Four of six met, and the two open ones are in the document too

Five seeds × five instruments = 25 measurements of 90,000 steps each. A result that holds for only one seed is not a result; that is why it is measured across seeds and the scatter is reported with it. The two highlighted rows are **not** met, and that is stated here, because a project that softens its own criteria after the fact no longer measures anything. Why they could not be solved is explained in the handbook: the force that keeps a price at fair value is the same one that makes its returns predictable.

These five pages

Page	What it covers
2	Starting up, and how the four views differ from each other
3	Who trades, and how orders become a price
4	The five ways of working, from “watch it once” to “test a claim”
5	Commands, where things are in the code, and what is open in the model

2 · Starting up and the four views

```
./start.sh -> http://127.0.0.1:8000
or double-click Market Town.app, which opens the browser by itself
```

The first start creates the virtual environment and installs the dependencies. After that the server fast-forwards to 110 trading days in the background. The validation panel needs 100 days of daily data before its figures mean anything. The bar at the top shows how far it has got. Pause and speed take effect immediately.

Four angles on the same run

One simulation runs. All four pages show it, but they answer different questions, and each can be reached from every other.

Page	Answers	What it is for
/ — the town	What is happening, and why?	A bright 3D diorama under white glass panels: every district a group of agents, every block height its capital, every road width this step's order flow, five blue towers the exchanges. None of it is decoration. On the left price, candles and order book; on the right a single agent with the calculation behind its last decision.
/sectors — analysis	Are the figures right?	Sector comparison, measured correlations, Central Bank with duration per sector, the validation panel, A/B comparisons across several seeds and the recorded runs.
/indices — indices	Who ran, when, and against what?	All indices live on one axis, normalised to the start of the window. For each index its level, fair-value index, volatility, drawdown, breadth, and the *measured* beta against the one set in the model.
/traders — traders	Who is trading, and how are they doing?	All 615 agents side by side instead of one after another: capital, cash, positions, leverage, result since the start of the run, forced sales. Sortable and filterable, with a jump back into the town.

The first look: what to watch in the first two minutes

- **The price against fair value** (market panel, left). The two lines rarely meet. How far apart they are is the most interesting single figure in the model, and the gap does not close by itself.
- **The ratio of momentum to value** in the Adaptive Quarter. It tips over in waves and then stays put for a while. That persistence is exactly what makes volatility cluster.
- **The red beacon and ring at the Leverage Tower.** They mean a fund is selling because it *must*, not because it chose to. This is the only place where an agent trades without deciding.
- **The order book** (third view in the market panel). Almost all of the depth sits right at the price; behind it there are only a few shares. That is exactly why a liquidity hole can open here. A model with a price equation would have none, because it would have nothing that could run out.

Keys and mouse in the town

Key	Effect	Key	Effect
Space	pause / resume	1 2 3	speed 1× / 4× / 16×
]	a single step	+ – 0	camera closer / further / reset
R B C	shock: calm, boom, crash	?	open the explanation
click	select a district (a tower: its market)	drag	turn; scroll zooms, right-drag pans

Speed is the most important control: at 1× a trading day takes about two minutes of real time, at 16× a few seconds. For anything that has to collect trading days, such as validation and experiments, set the clock to 16×.

3 · Who trades, and how a price is made

Seven rules. Each exists because the model was visibly wrong without it, not because it completes the list.

Type	Count	Rule	Why it is in the model
Value investor	300	buys below its own estimate of fair value	the only force pulling the price towards value
Trend follower	the same	extrapolates the recent move, ignores valuation	the feedback loop that produces bubbles
Noise trader	200	trades for reasons the model does not represent	background liquidity; otherwise there is no other side
ETF / savings plan	30	receives money on a schedule and buys at any price	price-insensitive demand
Leveraged fund	20	trades on credit, is forced to sell when under-collateralised	the cascade; carries stress between sectors
Market maker	60	quotes both sides at all times, leans against its inventory	the only standing liquidity
Issuer	5	sells new shares when they are expensive, buys back when cheap	the only source of new supply

Value investors and trend followers are the **same** agents. Every 20 steps they reconsider which of the two rules to run, based on what has worked lately and on what the majority is doing. This switching is the engine: the population tips into a trend regime or a value regime and stays there for a while.

How a price comes about

- **A step is neither a second nor a day.** 240 steps make one trading day with one daily candle. The statistical properties apply to *daily* returns; at step level microstructure dominates.
- **5 instruments trade at once**, each with its own order book: technology, energy, financials, health care, consumer. Their fair values share a common market factor. Correlation *emerges* from that; it is not set as a matrix.
- **News never moves the price.** It moves fair value and sentiment. Whether the price follows is up to the agents: the value investor sees a cheaper asset, the trend follower sees others selling, the ETF does not read the news at all.
- **The Central Bank does not trade.** It sets a rate that arrives through three channels: valuation (duration per sector), the leveraged funds' financing costs, and inflows into the savings plans.

The five sectors are deliberately different characters

Sector	Beta	Character
TECH Technology	1.40	high beta, bubble-prone; reacts most strongly to market news
ENRG Energy	0.90	low beta, but the highest idiosyncratic volatility
FINL Financials	1.20	rate-sensitive: a rate step moves fair value most here
HLTH Health care	0.70	defensive, calm, little movement of its own
CONS Consumer	0.80	defensive, close to the overall market

If they were five copies of each other, the measured correlation between them would not be a result but a tautology.

4 · How to work with it

Five ways of working, from “watch it once” to “test a claim”. Each builds on the one before.

Way	Answers	How it works
1 · Load a scenario	What does this model do under these conditions? (10 presets)	Town → footer **SCENARIO**. Calm market, Bubble formation, Leverage crash, ETF dominance, News shock, plus Dotcom bubble 1998–2000, Financial crisis 2008, COVID crash 2020, Meme stock 2021, Rate turn 2022. A scenario sets conditions and supplies headlines; **it does not prescribe a price path**. Whether the price falls 5% or 40% is the model's answer.
2 · Guided experiment	Does the mechanism really show? (7 of them)	Analysis → **Guided experiments**. Each one sets up the market, says in advance what to watch for, and afterwards checks by *measurement* whether it worked, even when it did not. Starting one switches to 16×; 45–200 trading days are recommended, and the panel says how many minutes that still is.
3 · Turn parameters	What does it depend on? (55 controls)	Analysis → right-hand column, or town → **CONTROLS**. A dot after the name means it only takes effect after a restart. Departures from the calibrated default are marked orange, and **defaults** brings everything back. Only there do the validation figures mean what the README says about them.
4 · A/B across seeds	Is the difference larger than chance?	Analysis → **A/B comparison**. Two configurations from the same seed, with a table of differences. What matters is **across 3 seeds**: the signal/noise column says whether the difference stands out from the scatter between seeds. Below 1 it does not, however clear a single chart looks.
5 · Record and replay	What exactly happened back then?	`python scripts/run.py --steps 60000 --name baseline`, then **REPLAY** in the town. A replay is *recomputed* from the seed, not read from curves. That is why the whole state is available at every position: order book, districts, agents. The slider moves the complete map.

> **The one rule that carries everything else**
> **A figure from one run is not a result.** Once, one seed here produced an autocorrelation of –0.002 with all six criteria met, and it could not be repeated; the mean across four seeds was –0.19 with a scatter of 0.11. Reporting that run as a success would have been exactly the trap this project's method warns against. So: measure across seeds, write the scatter down, and when in doubt quote `scripts/calibrate.py` rather than the live panel.

A typical session

- **Watch.** Start, set the speed to 4×, let the town run for a few minutes and read the “What is happening” panel alongside. Every statement in it comes from a figure the simulation holds at that moment.
- **Form a hunch.** For example: “Without market makers volatility is higher.” Check on **/indices** and **/traders** whether the hunch depends on figures you can actually get at.
- **Have it checked.** If there is a guided experiment for it, use that. It checks itself and also says when it did *not* work.
- **Otherwise measure it yourself.** A/B across three seeds, and only believe the rows whose signal/noise is above 1.
- **Keep it.** `scripts/run.py` records the run, with seed and timeline. It can be stepped through again later instead of only being looked at as a curve.

5 · Commands, code and what is open

Command	What for
`./start.sh`	start the interface
`python scripts/calibrate.py --runs 5 --steps 90000`	the validation: the figures on page 1
`python scripts/sweep.py ablate --steps 80000`	switch off each mechanism in turn and measure what is missing
`python scripts/sweep.py param --name n_passive --values 0,50,100`	sweep one parameter across a range of values
`python scripts/run.py --steps 120000 --name baseline`	a run without the interface, exported and recorded
`python docs/build_handbook.py`	regenerate the full handbook
`python -m pytest -q`	324 test functions: invariants, not price forecasts

Where things are in the code

File	Contents
`mes/orderbook.py`	matching engine, price-time priority: the foundation
`mes/agents.py`	the seven decision rules; all of the behaviour is here
`mes/simulation.py`	the loop: one `step()` is one market period
`mes/instruments.py`	the five sectors and the factor model behind their fair values
`mes/indices.py`	the index construction: chain-linked, with the ETF's weights
`mes/metrics.py`	the criteria from section 1, as executable checks
`mes/config.py`	all 55 parameters, declared once; the control panel is generated from them
`mes/city.py` · `traders.py`	the two places where state becomes display figures

What is open

No model is finished, and this one writes down where it stands:

- **Two of the six criteria are not met** (page 1). Four rounds of work made the other four robust and did not move these two. They hinge on the same trade-off.
- **The market makers' inventory limit is defined twice**: the agent works with five times what the engine lets through. Which of the two figures applies changes the book depth and with it every figure in the validation. Documented in `INTEGRATION.md`.
- **The cash check uses the best ask**, but a large order runs through several price levels. Agents that are not allowed to borrow can briefly do so anyway.
- **No dividends, no earnings.** Fair value is an exogenous process, not an earned result; buybacks are paid from a cash pile that never fills.

In full, with seven guided experiments, how the results were derived, and everything that was tried and did not work, in the handbook: `docs/Market-Ecology-Simulator-Handbook.pdf`. The method and the figures in detail: `README.md`.